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Modeling Informal Paratransit for Local Public Transport Route Planning

A MATSim application to Tricycle-for-Hire (TfH) services in Puerto Princesa City, Philippines



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Why Puerto Princesa needs new modeling tools

Three-wheelers fill public transport gaps across the Global South; planning tools do not fit them.

Regulation: Local Government Units issue tricycle franchises and set fares, and the Public Transport Modernization Program ties tricycles to the Local Public Transport Route Plan. Yet the Omnibus Franchising Guidelines set no passenger-per-hour-per-direction thresholds for tricycles; over half of 1,600 Local Government Units asked for a "Special" NOC.

Modeling: the four-step model aggregates demand, assumes fixed routes and schedules, and cannot anticipate emergent behavior such as "scavenging" or congestion from oversupply.

48.6%

of daily motorized trips are by tricycle in a city of 307,079 residents

0.4

km of road per km², with mountainous north and south

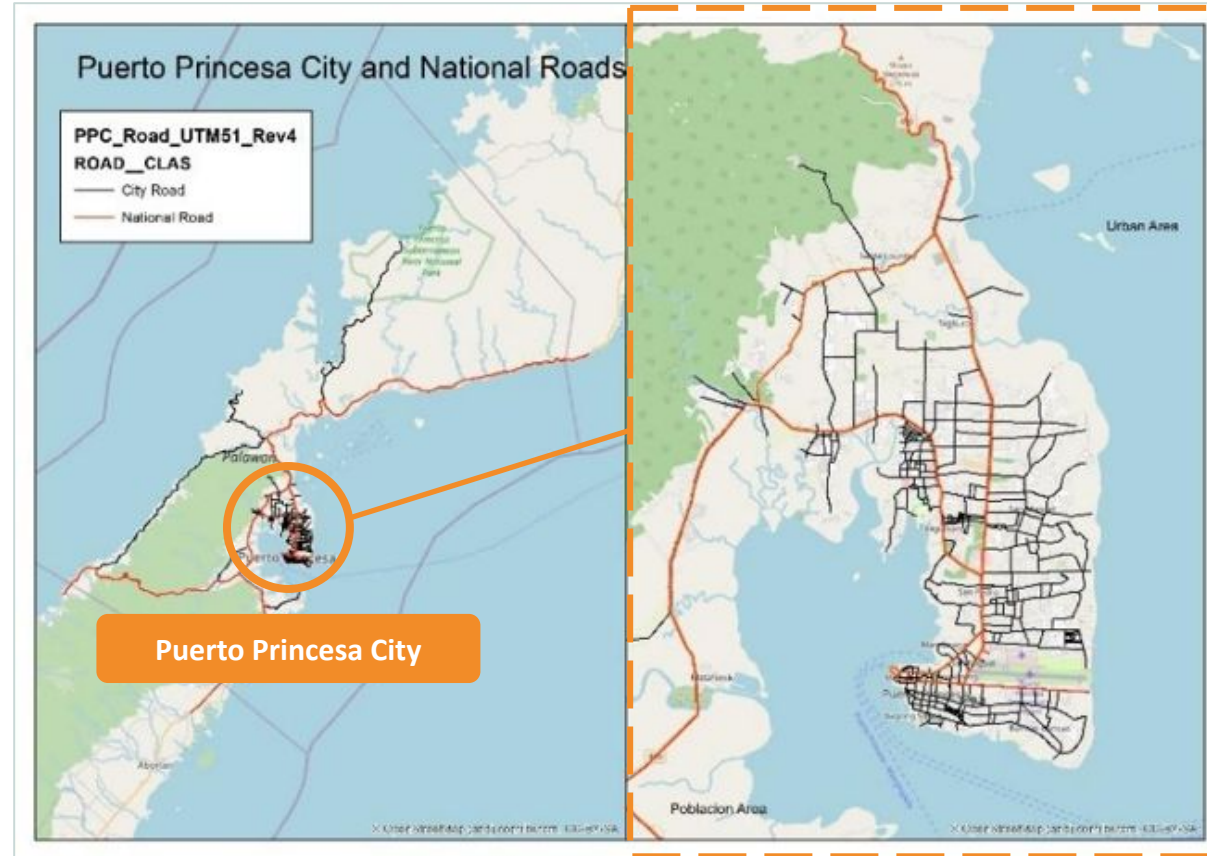
50–62%

of daily tricycle distance is dead mileage

Study area: Puerto Princesa City, Palawan



Location in the Philippines



Palawan (left) and the city road network (right): national roads in red, city roads in black

HUC

Highly urbanized city on Palawan

307,079

residents (2020 Census)

48.6%

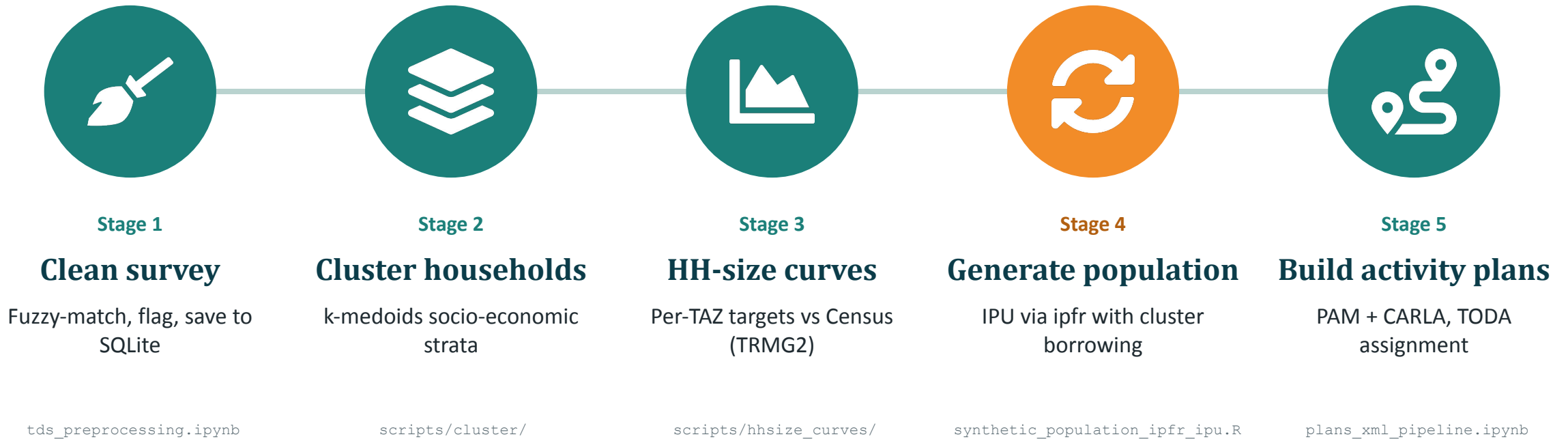
of daily trips by tricycle

0.4 km/km²

road density; mountainous north and south

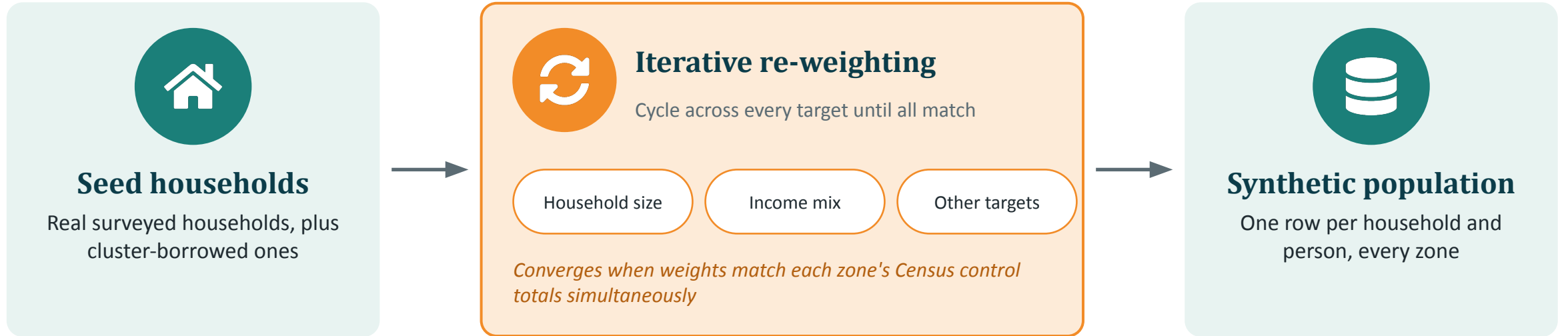
The five-stage pipeline

The pipeline fuses the Household Interview Survey (HIS) with the Census PUF to write a MATSim plans file for every resident.



Stage 4 is the centerpiece: the only step that actually creates the synthetic population.

IPU: the synthesis engine



Cluster borrowing

Where a zone has too few surveyed households to draw from, the pipeline borrows extra households from the most similar clusters elsewhere in the region, a method layered on top of standard IPU.

Scripts and outputs

```
synthetic_population_ipfr_ipu.R  
synthetic_population_ipfr_ipu-w-educ.R  
outputs/db/synthetic_population.db
```

IPU jointly fits Census PUF household and person marginals while preserving HIS joint distributions.

What makes tricycle-for-hire distinctive

Local Government Units issue tricycle franchises and regulate fares. Tricycles run with no fixed route or schedule inside a designated area, off national highways.



Fragmentation by operator

Not one fleet but dozens: in PPC, 20 TODAs, each with its own vehicles, terminals, and service level.



Territorial exclusivity

TODAs organize into separate areas of operation to resolve competition between them.



A dual trip regime

Shared rides along habitual corridors versus "special trips", where one passenger pays for the full vehicle.



Fixed boarding, flexible alighting

Riders board at a known terminal or shed but are dropped wherever they ask.

Why MATSim, and what stock DRT does not give you

MATSim's core loop is built for fixed-schedule mobility. DVRP adds fleets dispatched to real-time requests, and DRT on top of it is the natural starting point for on-demand tricycle trips.

Stock DRT assumes

One fleet, one dispatcher

One operating area

Every trip poolable

Symmetric stop-to-stop snapping



Tricycle reality

Dozens of independent associations

Licensed, often overlapping zones

Shared and exclusive-hire trips

Terminal boarding, free alighting

Hence two modes: trike_special (DRT, implemented) for full-vehicle hire; trike_shared (minibus contrib, in development) for corridor rides.

A two-mode architecture

trike_shared

IN DEVELOPMENT

Service	Quasi-fixed-route shared rides along habitual corridors
Contrib	minibus (org.matsim.contrib.minibus), after Neumann (2014)
Timescale	Across iterations: operators evolve routes via profit scoring and mutation
Maps to	Franchise system → franchises; stop areas → zoning; demand → hubs

trike_special

Service	Fully on-demand, point-to-point special trips
Contrib	DRT on DVRP, one fleet per TODA
Timescale	Within each mobsim: insertion-based dispatch
Maps to	Service-area polygons confine each fleet to its franchised zone

Why split? the two dominant service patterns follow different behavioral logic, so each uses the mechanism that fits it.

Status: trike_special runs on DRT/DVRP today; trike_shared on the minibus contrib is designed but not yet fully implemented.

trike_shared (in development): co-evolving operators

Neumann (2014)

Operators	Operators are agents that evolve routes, hours and fleets; bankrupt ones exit, new ones enter
Scoring	Each vehicle is scored as fare revenue minus distance, time and daily fixed costs
Franchise	A franchise system stops two operators from offering the same service
Demand	Activity-weighted demand knowledge guides where operators try new routes

Planned PPC mapping

Operators	Shared corridor service would evolve, making routes and fleet size outputs
Scoring	Profit-based scoring would test whether shared fares cover driver costs
Franchise	Franchise system and stop-provider areas would map onto franchises and zoning
Demand	Expected: organic hubs, like the terminals TODAs actually use

Why it fits: paratransit differs from DRT because it has no supervising controller (Neumann, 2014, §2.2), just like self-organizing TODAs.

Payoff: route rationalization in the Local Public Transport Route Plan: evolved fleets would indicate how many units each service area can support.

trike_special: dispatch, trip mix and fares



The central fix: real dispatch feasibility

Stock SelectiveInsertionSearch silently accepted every insertion regardless of its wait/travel-time promise. RealCostSelectiveInsertionSearch replaces the hardcoded zero-cost check and verifies the top 5 ranked candidates.



Shared vs special trips from field data

Special-trip shares for 9 surveyed TODAs range from 39.0% to 80.4%, with a 51.4% citywide fallback for the other 11. Until trike_shared is complete, each TODA's DRT mode draws shared or special per request.

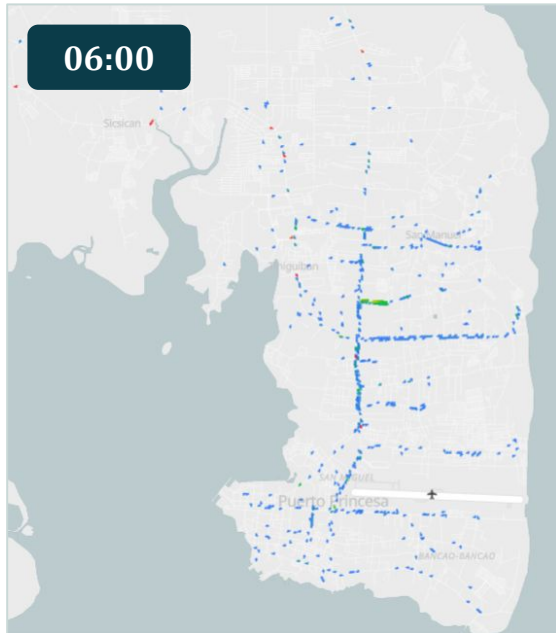


Fares for both tricycles and jeepneys

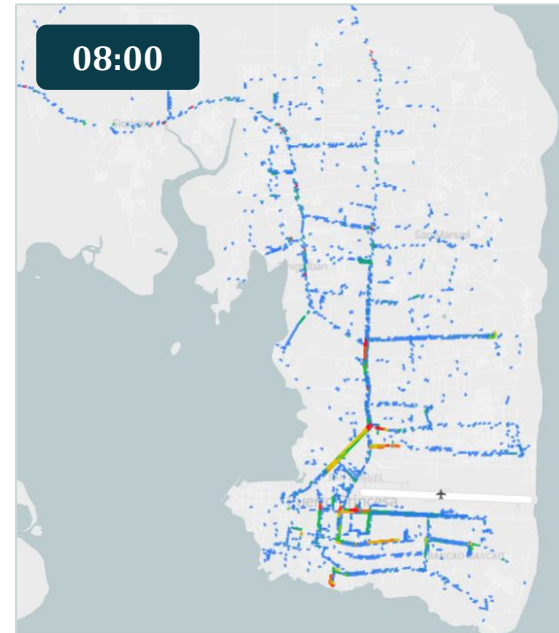
Tricycle shared ₱16–26 base + ₱2/km; special ₱60 base + ₱12/km. Jeepney ₱14 for the first 4 km + ₱2/km, charged per boarding. Bike was removed from mode choice so free teleported legs no longer absorb demand.

A simulated day: early morning to midday

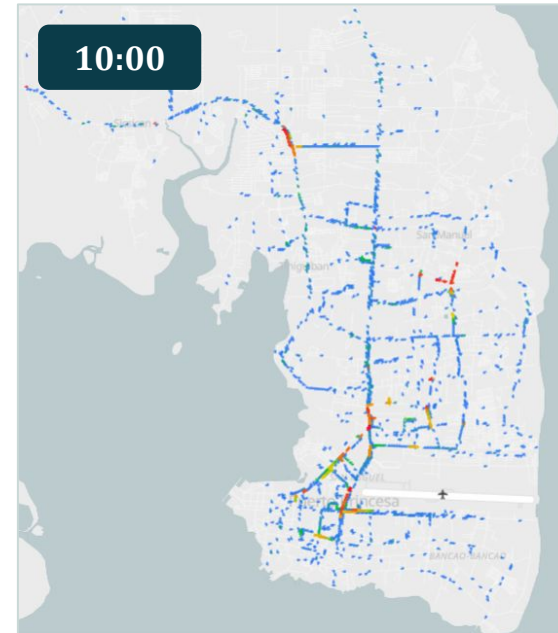
SimWrapper playback of the full scenario (299,085 agents) every two hours, using the current DRT-based TODA modes and jeepney pt.



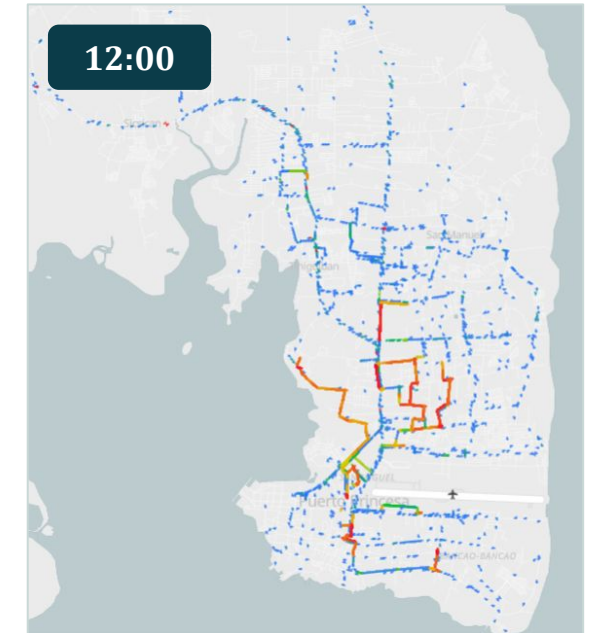
Sparse early trips along the north–south spine and a few cross streets.



Morning peak: dense movement on the spine and across the city proper.



Activity spreads over the street grid, busiest near the city proper.

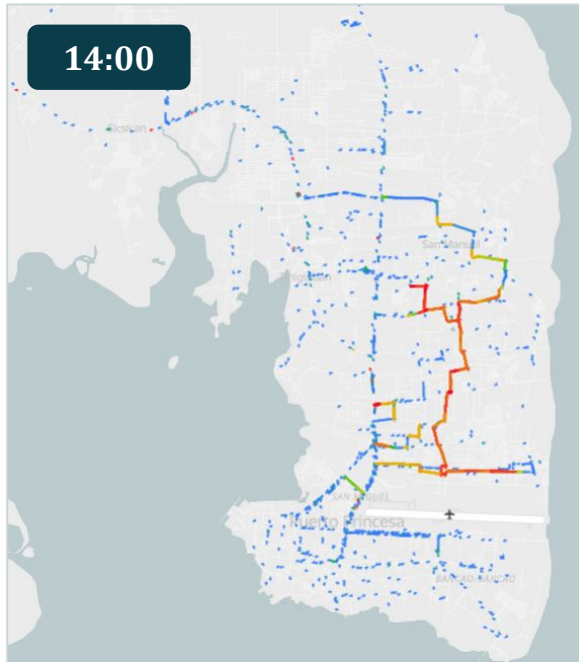


Midday: long continuous trails loop through the central barangays.

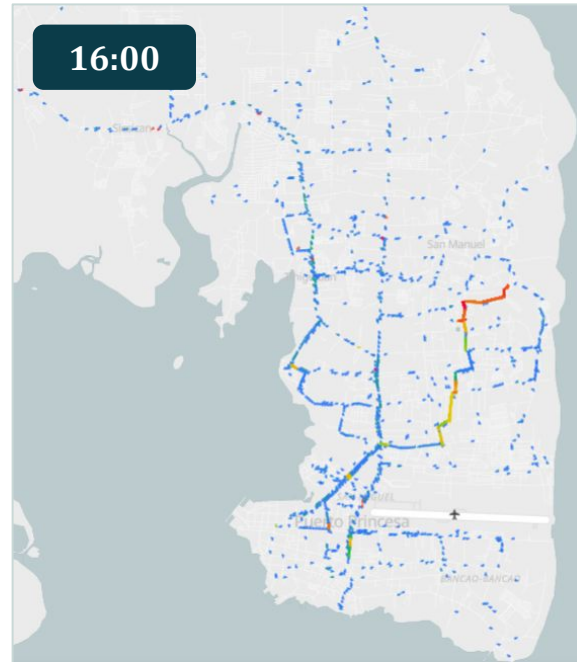
Reading the maps: movement concentrates on the north–south corridor and in the city proper, while the mountainous north and the airport area stay quiet all morning.

A simulated day: afternoon to evening

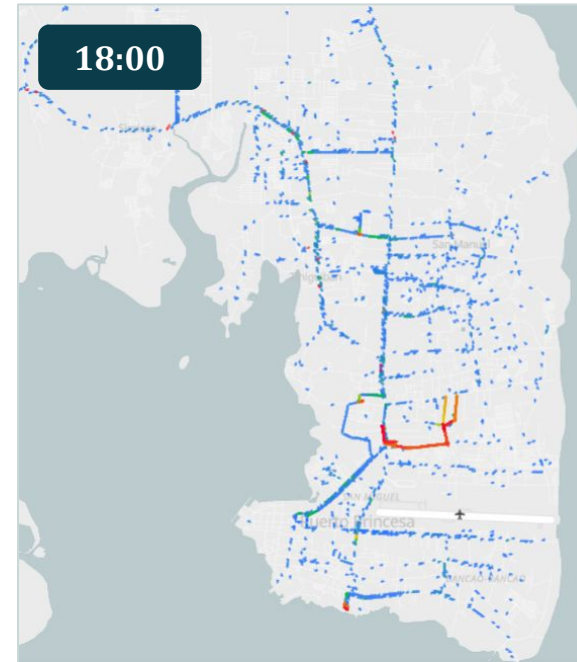
Same scenario and viewer settings; the afternoon shifts east, then flows back along the spine.



Afternoon trails extend east toward San Manuel.



Fewer long trails; movement returns to the main corridor.



Evening return: corridor traffic plus a busy central loop.

What the day shows

Corridor dependence

The north–south spine carries movement in every hour sampled.

Shifting hot spots

Busy areas move from the city proper (AM) to central and eastern barangays (PM).

Next check

Link these patterns to served vs rejected requests per TODA before drawing policy conclusions.

Limitations and way forward

1

Time-of-day subterminals

The pickup finder never receives departure time, so subterminal gating would need a full custom RoutingModule.

2

Passenger queueing

The inserter is scaffold only; direction-matching and FIFO-by-terminal queueing are future work.

3

trike_shared not yet complete

The minibus-based shared mode is designed but not fully implemented; current runs use the DRT modes.

4

Thinly surveyed TODAs

11 of 20 TODAs use the 51.4% citywide special-trip fallback rather than their own survey data.

5

Policy scenarios

Compare outputs against the 50-50 operating scheme and zoning scheme, then use them for scenario design.

Future improvements from Lu (2025)

Lu (2025)

Ride time	Maximum ride duration can be set explicitly in request insertion
Rebalancing	Min-cost-flow, adaptive real-time, feedforward and plus-one strategies share one interface
Pre-booking	A rolling horizon serves the same demand with fewer vehicles and shorter booking lead times
Estimation	A DRT estimator fitted to real data cuts runtime and run-to-run variation

Planned PPC improvement

Ride time	Set explicit ride-time promises alongside the RealCost check
Rebalancing	Test against return-to-terminal to cut the 50–62% dead miles
Pre-booking	Model booked special trips, including school runs
Estimation	Fit to tricycle IoT data for faster, stable full-scale runs

Why it fits: these are MATSim DRT/DVRP extensions, so they plug into the same override pattern `RunDrt.java` already uses.

Addresses: dead mileage, calibration, and runtime at city scale.

CONCLUSION

Contribution and next steps



Population

HIS fused with the Census PUF through IPU



Two modes

trike_special on DRT/DVRP now;
trike_shared on minibus next



Translation

Real TfH features mapped onto MATSim mechanisms

Next: complete trike_shared on the minibus contrib, benchmark against the city's 50-50 operating scheme and zoning scheme, then design route-rationalization scenarios for the Local Public Transport Route Plan.

MATSim can represent tricycle service as a complex adaptive system where passengers and operators interact.

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